

WINTER HAVEN, FL, USA

WMO: 747931

Lat: 28.062N

Lon: 81.754W

Elev: 146

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 12876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.8	43.0	20.9	15.8	48.2	26.8	20.9	52.8	21.3	63.1	19.2	61.7	7.6	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	94.5	75.5	93.1	75.3	91.6	75.1	79.0	87.0	78.3	86.3	77.9	85.9	6.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.4	81.5	76.4	138.5	81.1	75.6	134.9	80.8	42.5	86.9	41.8	86.7	41.2	86.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.8	15.2	DB	32.2	97.4	3.9	1.3	29.4	98.3	27.1	99.1	24.9	99.8	22.1	100.8	
			WB	28.6	80.2	3.9	1.0	25.8	80.9	23.6	81.5	21.4	82.1	18.6	82.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.8	61.0	64.6	67.9	73.5	78.5	82.0	83.1	83.3	81.8	76.3	68.6	64.7
	DBStd	9.55	8.14	7.88	6.45	4.76	3.74	2.67	2.35	2.22	2.26	5.43	6.45	7.92
	HDD50	25	15	4	0	0	0	0	0	0	0	0	0	6
	HDD65	473	171	96	49	6	0	0	0	0	0	6	40	105
	CDD50	8726	355	414	555	705	884	961	1028	1034	954	816	557	463
	CDD65	3700	47	86	138	261	420	511	563	569	504	357	147	97
	CDH74	32569	250	543	953	2091	3771	4829	5598	5734	4617	2826	862	495
	CDH80	12379	15	93	202	664	1601	2079	2345	2364	1829	958	162	67
(14) Wind	WSAvg	7.0	7.2	7.1	8.1	8.0	7.6	6.3	5.4	5.8	6.5	7.2	7.2	7.0
(15) Precipitation	PrecAvg	52.0	2.8	2.4	3.6	2.8	3.1	8.7	7.8	8.3	6.5	2.5	1.6	2.8
	PrecMax	71.9	6.1	4.4	7.8	6.5	8.8	13.6	11.8	11.7	14.6	6.2	3.9	12.3
	PrecMin	39.2	0.6	0.6	0.3	0.4	1.2	3.4	3.5	4.6	2.5	0.1	0.1	0.6
	PrecStd	8.6	1.6	1.3	2.6	1.6	2.1	3.2	2.2	2.0	3.5	1.7	1.1	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.9	85.5	86.6	90.5	95.0	96.5	95.9	95.6	94.1	92.0	87.6	84.1
		MCWB	68.9	69.9	69.3	71.3	71.7	75.5	76.1	77.0	76.2	75.5	74.0	72.0
	2%	DB	79.4	82.9	84.4	88.2	92.2	93.8	93.8	93.6	92.0	89.9	84.0	81.8
		MCWB	67.0	68.1	68.0	70.0	71.5	74.7	76.1	76.5	75.4	75.1	71.7	70.2
	5%	DB	77.1	80.7	82.3	86.3	90.4	92.0	92.3	92.4	90.7	87.8	81.5	79.6
		MCWB	65.6	66.9	66.7	68.9	71.1	74.7	76.1	76.5	75.2	73.9	69.1	69.0
	10%	DB	73.8	77.6	80.1	84.1	88.3	90.2	90.6	90.7	89.2	85.4	79.2	76.8
		MCWB	64.7	65.8	65.6	67.7	70.8	74.6	76.1	76.5	75.2	71.7	67.4	67.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	72.7	73.2	75.0	76.4	79.2	80.0	80.2	79.3	78.3	75.6	74.1
		MCDB	78.2	80.5	79.8	83.0	84.5	87.5	87.7	88.5	86.8	85.7	82.8	80.4
	2%	WB	69.8	70.8	71.5	73.2	75.3	78.0	78.5	79.1	78.3	77.2	74.1	72.6
		MCDB	75.2	77.7	78.5	82.0	83.9	86.8	86.7	87.1	85.8	84.8	80.6	78.4
	5%	WB	68.3	69.3	70.2	72.0	74.4	77.2	78.1	78.4	77.7	76.2	72.5	71.1
		MCDB	73.7	76.2	77.0	80.7	83.0	86.1	86.3	86.3	85.1	83.3	78.1	75.9
	10%	WB	66.4	67.9	68.6	70.8	73.4	76.5	77.4	77.9	77.0	75.3	70.6	69.6
		MCDB	72.0	75.0	75.8	79.4	82.1	84.8	85.6	85.8	84.3	81.7	76.0	74.5
(35) Mean Daily Temperature Range	5% DB	MDBR	19.8	20.2	20.3	20.3	19.5	16.9	15.9	15.6	15.7	17.7	18.8	18.6
		MCDBR	20.6	21.0	21.0	21.1	21.3	19.1	17.5	17.4	17.0	17.2	18.6	18.6
		MCWBR	9.8	8.5	7.4	6.6	5.4	5.1	4.7	4.7	4.7	5.3	7.7	8.5
	5% WB	MCDBR	17.7	18.4	17.8	18.6	17.8	16.4	15.6	15.5	15.0	15.3	16.1	16.3
		MCWBR	10.4	9.0	8.5	7.6	5.7	5.3	5.2	5.2	4.9	5.1	7.8	9.0
(40) Clear-Sky Solar Irradiance	taub		0.339	0.351	0.368	0.385	0.422	0.450	0.472	0.457	0.421	0.394	0.357	0.342
	taud		2.512	2.489	2.434	2.364	2.308	2.305	2.241	2.305	2.390	2.431	2.486	2.530
	Ebn at Noon		287	292	293	289	277	267	261	265	272	274	278	281
	Edh at Noon		29	32	36	40	42	42	44	42	37	34	30	27
(44) All-Sky Solar Radiation	RadAvg		1114	1327	1664	1967	2048	1824	1791	1722	1567	1445	1189	1020
	RadStd		81	80	124	97	134	158	115	113	87	95	60	79

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-7	-81	+284	+202

Nomenclature: See separate page